

DSC 40B Discussion 5 Worksheet

Friday, May 1, 2026

Imagine you are a tutor, and you need to explain something to someone who has only a vague understanding of what's going on. Do your best to get the point across. **ALTERNATE**.

If you're not sure, ask your peer if they know how to explain it, then REPEAT the explanation back. Remember, the point of the exercise is to **vocalize** your thoughts.

If you are both unsure, make sure to come to/tutor for help! Do not use ChatGPT or any other LLM! The point of discussion is to prepare you for the exams.

Have one person in the pair open the Discussion 1 Gradescope assignment. After both you **and** your partner finish a problem **and explain it**, answer the corresponding multiple choice/short answer questions for both versions.. You will receive instant feedback on the questions, whether you are correct or not. If your answer is incorrect, please go back to the question to understand it and feel free to ask tutors about anything you are confused about. Once you have finished the entire worksheet or discussion is ending, turn in your assignment and add your partner to the submission.

Quick Icebreaker

Introduce yourself to your partner by sharing name, year, major, etc. What is your favorite off-campus food spot in San Diego? (Do not spend more than 2 minutes!)

Questions

Question 1:

First Person:

```
def mystery1(arr):
    count = 0
    for x in arr:
        y = random.random()
        if x % 2 == 0:
            count += 1
            print(y)
    return count
```

For the code snippet above, is it more appropriate to analyze its time complexity with average or expected time complexity. Explain your reasoning.

Second Person:

```
def mystery2(arr):
```

```

count = 0
for x in arr:
    y = random.random()
    count += 1
    if random.random() * 2 < 1:
        return count
return count

```

For the code snippet above, is it more appropriate to analyze its time complexity with average or expected time complexity. Explain your reasoning.

Question 2:

Second Person:

Permutation sort is a sorting algorithm that deterministically cycles through all $n!$ ways to arrange the elements of an input list of size n , and terminates once it has found a permutation in which the list is sorted. Bogosort is a sorting algorithm that randomly shuffles the elements of an input list and terminates once the list is sorted. Match each of these algorithms with the type of runtime analysis that makes the most sense to perform.

- ☐ Expected for both
- ☐ Expected for permutation sort and average case for bogosort
- ☐ Average case for permutation sort and expected for bogosort
- ☐ Average case for both

First Person:

For each of the following algorithms, determine whether it makes sense to compute the average case or the expected time complexity:

- Performing a binary search for the number 53 on a sorted list
- Inserting n elements into an empty BST (not self balancing) and then searching for 53

- ☐ Expected for both
- ☐ Expected for binary search and average case for BST
- ☐ Average case for binary search and expected for BST
- ☐ Average case for both

Question 3:

First Person:

Given `arr` is an arbitrary list with random integers, you perform randomized quickselect (where the pivot index is selected randomly) on `arr`. To best analyze the asymptotic time complexity, should you compute average or expected time complexity? Explain your reasoning.

- ☐ Average

☐ Expected

Second Person:

Given `arr` is an arbitrary list with random integers, you perform fixed quickselect (where the pivot index is always the first element) on `arr`. To best analyze the asymptotic time complexity, should you compute average or expected time complexity? Explain your reasoning.

- ☐ Average
☐ Expected

Question 4:

Second Person:

True or False? Quicksort always runs in $O(n \log n)$ time because each partition step takes linear time. Explain your reasoning.

- ☐ True
☐ False

First Person:

True or False: In randomized quicksort, the input array does not need to be randomly ordered for the algorithm to have good expected runtime.? Explain your reasoning.

- ☐ True
☐ False

Question 5:

Example Quicksort Tracing:

[2, 4, 1, 3]

→ [2, 1, 3, 4]

→ [1, 2, 3, 4]

First Person:

For the list below, run fixed quicksort with the pivot as the last element and in-place partitioning (Approach #2 in lecture) on it and trace the list in the same way as the example above (after each “layer” of partitions).

[4, 1, 6, 2, 7, 3, 5]

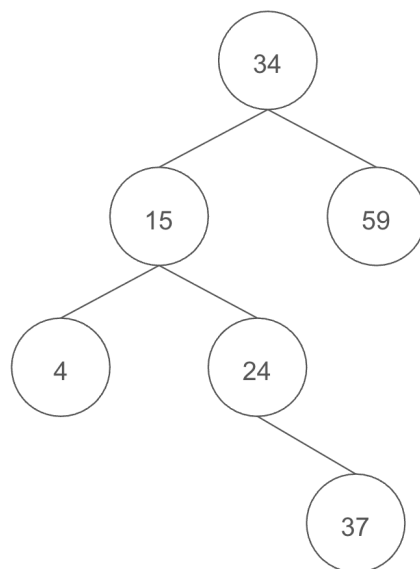
Second Person:

For the list below, run fixed quicksort with the pivot as the last element and in-place partitioning (Approach #2 in lecture) on it and trace the list in the same way as the example above (after each “layer” of partitions).

[6, 2, 7, 1, 5, 3, 4]

Question 6:

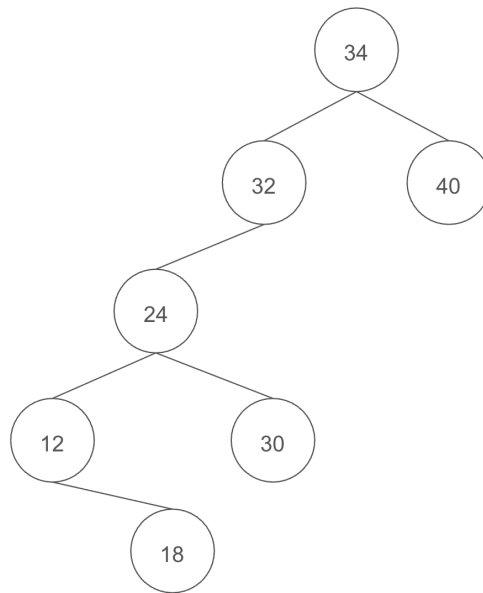
Second Person:



Is the tree picture above a valid binary search tree? Explain your reasoning.

- ☐ Valid
- ☐ Not Valid

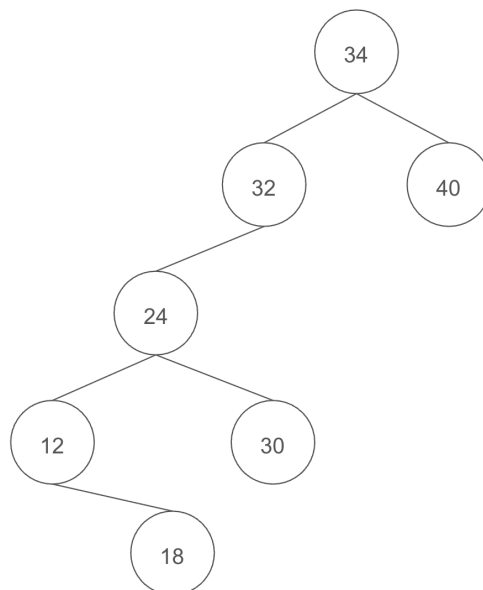
First Person:



Is the tree picture above a valid binary search tree? Explain your reasoning.

- ☐ Valid
- ☐ Not Valid

Question 7:



First Person:

Given the binary search tree above, insert the following sequence of numbers and draw the final tree.

[28, 39, 36, 10]

Second Person:

Given the binary search tree above, insert the following sequence of numbers and draw the final tree.

[33, 20, 19, 42]

Question 8:

Second Person:

Is the hash function pictured below a good hash function? Explain your reasoning.

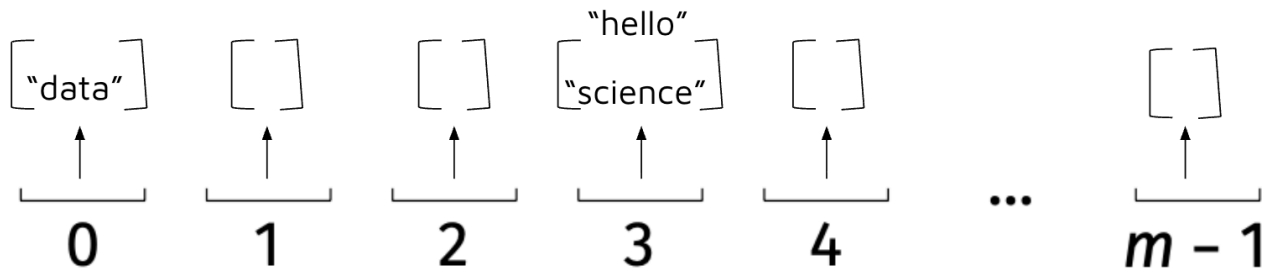
```
def hash_A(x, m): # m is table size
    return int(m * x + 8) % 10
```

First Person:

Is the hash function pictured below a good hash function? Explain your reasoning.

```
def hash_B(x, m): # m is table size
    return int(random.random() * x - 7) % m
```

Question 9:



Hash table after hashing and inserting the sequence ["data", "science", "hello"].

You are given an empty hash table initialized with $m=5$ buckets that handles collisions with separate chaining (as pictured above) and the below hash function.

```
def hash_func(x, m): # m is table size (5)
    return (3 * x + 8) % m
```

First Person:

Draw the hash table in the same way as above after inserting the following sequence of numbers.

[4, 6, 1, 9, 3, 8, 0, 2, 10]

Second Person:

Draw the hash table in the same way as above after inserting the following sequence of numbers.

[1, 6, 11, 16, 21, 2, 7, 12]